

Concept Note - Team-2

01) Understanding the Problem:

01.1) PS-02: Climate Stress Early Warning for Biological Intervention

- A. **Climate stress can damage crops before visible symptoms appear:** Abiotic stresses such as heatwaves, droughts and unseasonal rainfall can start damaging a crop long before the farmer can see symptoms like wilting. The problem becomes even more serious because the same weather event can affect a crop very differently at different growth stages. A high temperature during vegetative growth may cause limited damage, while the same temperature during flowering can severely affect the crop.
- B. **Farmers lack a field-level, phenology-aware warning for preventive action:** Today, farmers have access to broad weather forecasts, but they do not have a field-level indication of when drought or heat stress is likely to affect their crop or how much time they have to protect it. The critical gap is therefore the lack of a phenology-aware, early warning system that converts upcoming weather into a clear window for preventive action.

01.2) PS-03: CropFit - Personalised Biological Product Advisor

- A. **Choosing the right biological input requires more than selecting a product:** Getting the best results from biological inputs requires more than simply choosing a product but involves selecting the right formulation, dosage, application method and timing depending on the crop, its growth stage and the conditions of the individual field. Today, farmers often have to rely on their own experience, trial-and-error or informal retailer recommendations because there is no clear, personalized guidance for their specific situation.
- B. **The same product or dosage may not work equally for every farmer:** Even two neighbouring farmers growing the same crop can have different soil conditions and different crop needs, meaning a product or dosage that works well for one farmer may not necessarily be the right choice for another. This creates a practical gap where under-application may make the intervention ineffective, while over-application increases costs without providing proportional protection.

01.3) PS-07: Yield Impact Predictor - Did the Biological Make a Difference?

- A. **Farmers cannot clearly measure the true impact of a biological input:** Measuring the true economic value of a biological input is difficult because every crop cycle is influenced by changing weather, soil conditions and management practices. Since final yield reflects all of these factors together, farmers struggle to understand how much of the yield or protection was a result of the product they applied.

- B. **Farmers lack a clear connection between product impact and profitability:** Farmers need to make decisions based on what their crop will ultimately be worth in the local Mandi market, but they often lack a clear connection between expected yield impact, current market prices and the cost of the intervention. Without this combination of verifiable impact attribution and local economic context, farmers find it difficult to judge whether an intervention was truly profitable, creating a trust gap that makes them hesitant to move away from familiar chemical solutions.

01.4) PS-04: Multilingual Biological Advisory Chatbot for LowLiteracy Farmers:

- A. **Agronomic insights are only useful when farmers can understand and act on them:** In smaller farming regions, many farmers rely on regional languages and have varying levels of digital and textual literacy, while some may not be able to read at all and communicate only through their native language. Yet, most existing advisory tools depend on technical, text-heavy interfaces, creating a communication gap where important predictive warnings may not reach the farmer in a form they can use.
- B. **Critical information needs to be delivered in a form accessible to every farmer:** Our idea addresses this by making the application's key recommendations, alerts and guidance available in speech, in the farmer's preferred language, so that critical information can translate into timely and informed action in the field.

01.5) Real-World Context:

A farmer navigating a growing season faces a chain of interconnected uncertainties. When unseasonal heat, drought or other stress hits during a sensitive growth stage like flowering, damage can begin long before visible symptoms appear, while the farmer may have no clear way to know when the stress is coming or how much time is left to protect the crop. When looking for a solution, he often has to rely on experience, word of mouth or generic recommendations, but these recommendations are the same for everyone and do not account for differences in soil, sowing time, crop stage or field conditions, making it difficult for the farmer to trust that the advice is right for his field and also possibly affecting his crop yield. At the same time, many farmers struggle to understand technical advisories because they are not provided in their native or preferred language, and some may not be comfortable reading at all. At harvest, when yields vary despite significant input costs, the farmer still cannot clearly determine whether the treatment actually protected his crop or whether the outcome was mainly driven by weather and other field conditions. Without early visibility of stress, personalised guidance, communication in a language he understands, and clear proof of value, every seasonal investment remains an uncertain gamble.

02) Proposed Solution:

02.1) PS-02: Climate Stress Early Warning for Biological Intervention:

- A. **Phenology-aware stress prediction:** Once the farmer registers his field location and crop, the application combines upcoming weather conditions with a Growing Degree Days (GDD) model to understand exactly where the crop is in its growth cycle - from early root establishment to the more vulnerable flowering and fruiting stages. Since stress at an early stage can affect the crop's performance later, the system considers both how severe the upcoming stress is and how sensitive the crop is at that particular stage.
- B. **Biological Activation Countdown:** Instead of simply telling the farmer that a heatwave or drought is coming, our solution provides a Biological Activation Countdown showing how quickly the opportunity for protection is declining. For example, it explains the difference in yield protection between acting on Day 1 versus the decline on waiting until Day 3. This gives the farmer the crucial lead time to understand the risk, arrange the right intervention and act before the stress causes irreversible damage.

02.2) PS-03: Personalised Product & Economic Recommendation

- A. **Personalised product recommendation:** Instead of giving every farmer the same product recommendation, the application looks at the specific conditions of that farmer's field - including soil characteristics, crop, sowing date, growth stage, expected weather and other field-level factors. It matches the expected stress with the most suitable biological product and determines what to apply, how much to apply, when to apply it and where/how it should be applied, along with a confidence score.
- B. **Economic decision support:** It also considers the product cost, labour cost and seasonal local Mandi prices to understand whether the expected yield protection is actually worth the investment. This helps the farmer avoid unnecessary applications and make a personalised, financially informed decision rather than relying only on experience or generic recommendations.

02.3) PS-07: Yield & Economic Planning

- A. **Field-specific product and application recommendation:** Instead of giving every farmer the same product recommendation, the application looks at the specific conditions of that farmer's field - including soil characteristics, crop, sowing date, growth stage, expected weather and other field-level factors. We believe that even two neighbouring farmers growing the same crop can have very different outcomes because their soil, sowing time and farm practices are different. Our application therefore matches the expected stress with the most suitable biological product and determines what to apply, how much to apply, when to apply it and where/how it should be applied, along with a confidence score.
- B. **Economic decision support:** It also considers the product cost, labour cost and seasonal local Mandi prices to understand whether the expected yield protection is actually worth the investment. This helps the farmer avoid unnecessary applications and make a

personalised, financially informed decision rather than relying only on experience or generic recommendations.

02.4) PS-04: Voice-First & Accessible Advisory

- A. **Voice-first, multimodal interaction:** Predictive insights are only useful if the farmer can understand them and act on them. We propose a voice-first, multimodal conversational interface built around this idea. We use Google Cloud Speech-to-Text and Text-to-Speech to support regional languages such as Hindi, Marathi, Tamil and Telugu, while Google Gemini 1.5 Pro helps understand farmer intent and analyse crop images. Farmers can ask questions through voice notes, share crop photos, or simply listen to important recommendations and alerts from the application.
- B. **Simple, actionable advisories:** The system then provides short, jargon-free audio advisories in the farmer's preferred language, helping turn complex technical recommendations into something the farmer can confidently understand and act on in the field.

Rationale Behind the Proposed Solutions

- **PS-02:** Biological inputs operate via preventive physiological modes of action (e.g., osmotic balance, cellular stabilization); predicting stress with growth-stage lead time is the only way to ensure input application occurs before irreversible tissue damage.
- **PS-03:** Agronomic efficacy and economic ROI depend non-linearly on crop phenology and dosage; combining biological modes of action with live market prices ensures farmers maximize net financial return rather than just volume.
- **PS-07:** Farmers default to familiar chemicals because seasonal weather noise obscures biological product impact; causal inference provides rigorous, unconfounded proof of yield preservation required to establish trust.
- **PS-04:** Predictive intelligence is ineffective if smallholders cannot comprehend it; voice-first regional interaction aligns directly with the communication habits of rural smallholders

03) The AI/ML Component: Hybrid Architecture:

Our system uses a **hybrid AI/ML architecture** that combines mechanistic agricultural models, machine learning, causal inference, and rule-based decision logic. Instead of relying entirely on a black-box model, each layer contributes a specific type of reasoning and can be evaluated independently.

03.1) Hybrid Pipeline:

Phase	Component	Role
1. Zero-Touch Onboarding	Gemini + Speech-to-Text	Converts vernacular farmer voice into structured inputs such as crop, sowing date and location.
2. Data Ingestion	Meteoblue, CE Hub, Agmarknet	Collects weather, soil and market information through APIs.
3. Mechanistic Layer	GDD + FAO Penman-Monteith / VPD	Converts raw environmental data into biologically meaningful growth and stress indicators.
4. ML & Causal Layer	XGBoost + Double ML	Estimates crop-stress risk and the counterfactual yield without intervention.
5. Decision Layer	CropFit + Economic Optimizer	Combines agronomic constraints with economic conditions to determine product and dosage.
6. Multimodal Output	Google TTS	Converts the final recommendation into a local-language voice advisory.

The architecture therefore follows a simple principle:

Raw Inputs → Biological Features → Risk & Efficacy Estimation → Agronomic + Economic Decision → Farmer Advisory

03.2) Core Intelligence Engines:

1. GDD Phenology Engine: Mechanistic

$$GDD = \max(0, \frac{T_{max} + T_{min}}{2} - T_{base}) \quad \text{(Formula 3.1)}$$

GDD estimates the crop's current growth stage from accumulated heat units rather than calendar dates, reducing dependence on large labelled datasets.

2. Atmospheric & Water-Balance Engine: Mechanistic

The system uses ET and VPD to represent atmospheric drying pressure and water stress. Pre-calculated FAO Penman-Monteith ET from the weather API reduces dependence on additional local sensing infrastructure.

3. Compound Stress Engine: Mechanistic + ML

Heat Stress Index (HSI) and Drought Stress Index (DSI) are combined with growth-stage weighting to capture compound stress:

$$\text{Compound Stress} = \text{HSI} \times \text{DSI} \times \text{Synergy} \quad (\text{Formula 3.2})$$

This allows simultaneous heat and drought conditions to have a stronger effect than either stress considered independently.

4. Weather: Yield Model: Machine Learning

An XGBoost-based model uses engineered weather and crop features such as accumulated GDD, heat-stress days, rainfall and dry-spell characteristics to model non-linear weather–yield relationships while retaining feature-level interpretability.

5. Confidence Scoring: Decision Logic

The recommendation confidence combines forecast reliability, product–condition matching and supporting evidence:

$$\text{Score} = 0.40(\text{Forecast}) + 0.35(\text{Match}) + 0.25(\text{Evidence}) \quad (\text{Formula 3.3})$$

This provides a transparent confidence layer rather than presenting every recommendation with the same certainty.

03.3) Multimodal Fallback:

The primary onboarding flow uses vernacular voice input. If the farmer cannot provide a reliable sowing date, the architecture includes a **Gemini Vision fallback** that can use a crop image to estimate the current growth stage, allowing the downstream pipeline to continue.

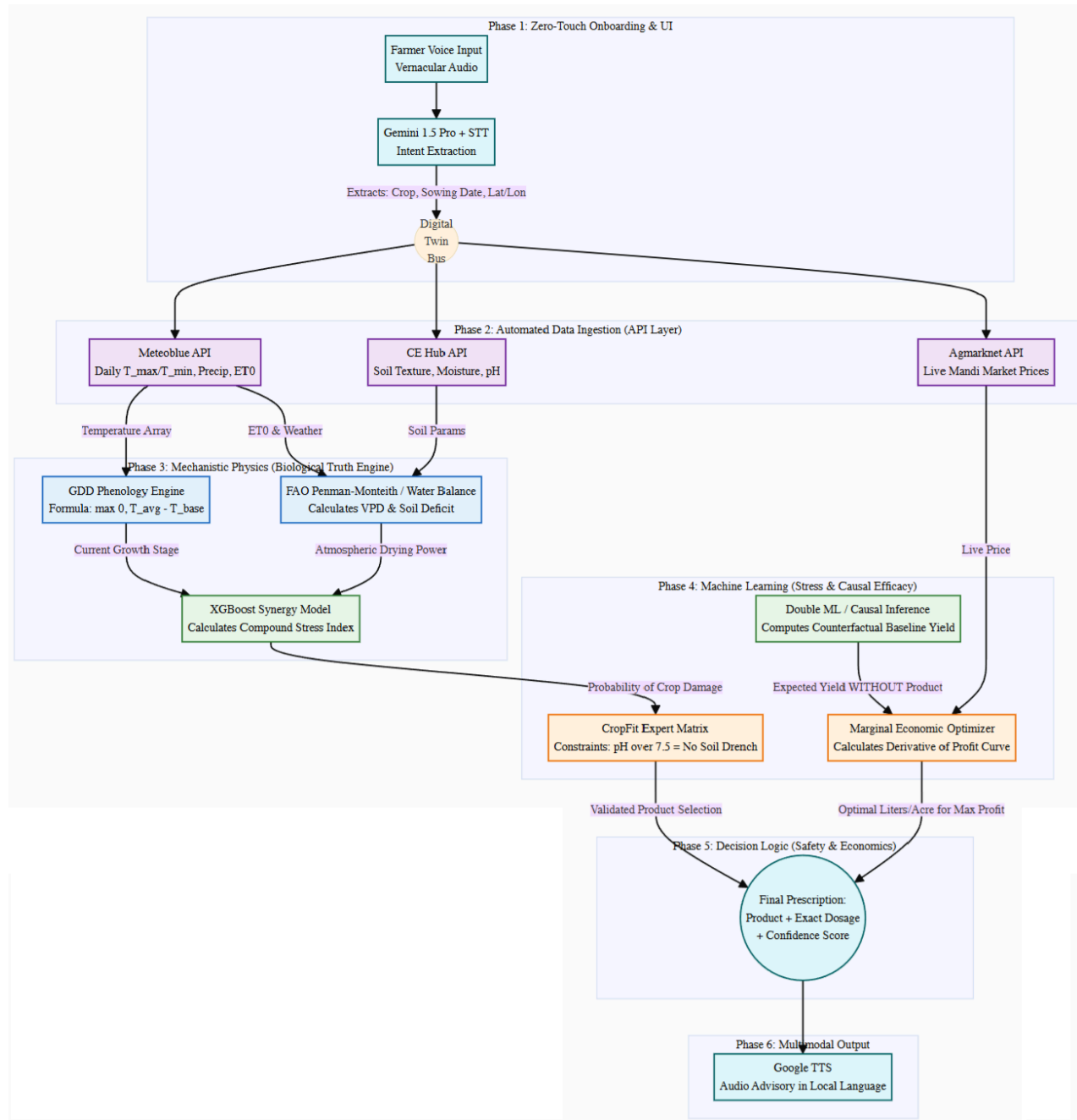


Figure 1. Proposed System Hybrid Architecture

03.4) Architectural Model Selection & Comparative Evaluation :

Rather than applying machine learning to every component, our solution selects the simplest suitable approach for each problem. We compared multiple candidate architectures across four core engines, considering **accuracy, data requirements, interpretability, safety and MVP feasibility**.

The architecture combines mechanistic modelling, machine learning, causal inference and deterministic decision logic into a single pipeline.

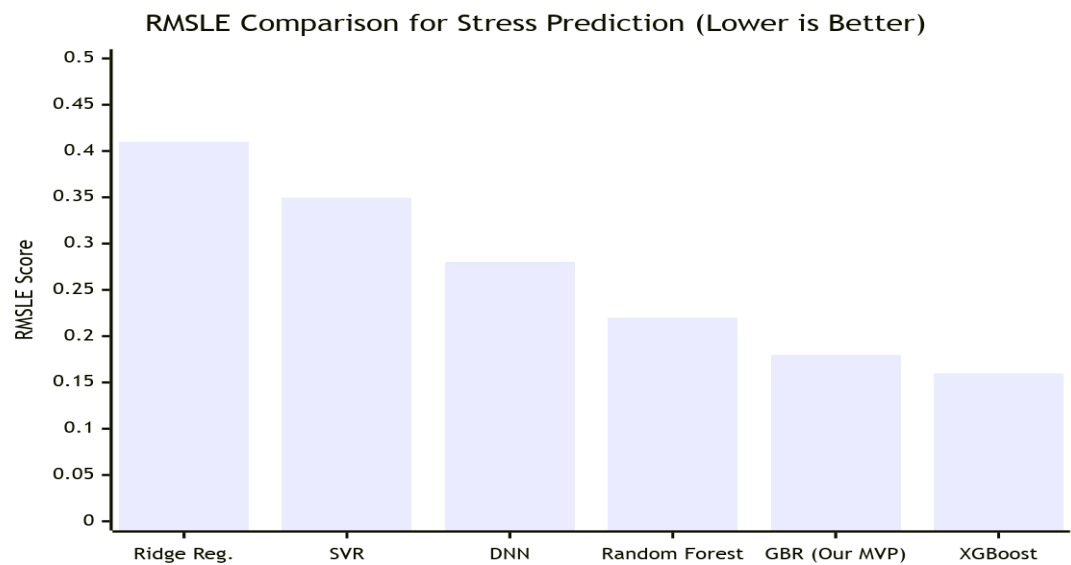


Figure 2. RMSLE Comparison

Engine	Selected Approach	Alternatives Considered	Why We Selected It
<i>Stress Prediction</i>	Gradient Boosting / XGBoost	Ridge, SVR, Random Forest, DNN	Captures non-linear weather interactions while remaining suitable for tabular data and interpretable through feature importance.
<i>Growth Stage</i>	GDD Phenology	Random Forest, CNN, LSTM, YOLO, HMM	Uses accumulated heat units directly, requiring no large labelled dataset or farmer-provided imagery.
<i>Product Recommendation</i>	CropFit Expert Matrix	KNN, LLM, Decision Tree, Collaborative Filtering, RL	Known agronomic safety constraints are better enforced deterministically than learned probabilistically.
<i>Dosage Optimisation</i>	Marginal Economic Optimisation	DQN, MLP, Genetic Algorithm, Bayesian Optimisation, Polynomial Regression	The problem is a one-dimensional economic optimisation, making direct mathematical optimisation simpler and more reliable.

The detailed comparison in the source evaluates these four engines against five alternatives each, covering 20 alternative model choices in total.

The mechanistic layer generates biologically meaningful features such as *growth stage*, *ET*, *VPD* and *compound stress*; the ML layer handles non-linear weather–yield relationships; and the decision layer combines agronomic constraints with economic optimisation. This is what makes the architecture hybrid rather than simply “**ML-based.**”

04) Use of Datasets & Resources:

The concept follows the HACK CORE 2026 guidance: organizer-provided datasets and APIs are the primary foundation.

04.1) Primary organizer-provided resources:

Resource	Purpose
Meteoblue Dataset API	Historical + forecast weather, soil, evapotranspiration and vegetation information for stress, weather and yield modules.
Syngenta CE Hub APIs	Used to cross-check derived stress indicators and support agronomic logic
ANNAM.AI	Climate datasets Ground-truth climate validation.
Syngenta product-efficacy data	Product recommendation and intervention-efficacy logic for PS-03.

04.2) Supplementary open resources:

Resource	Purpose
data.gov.in	District-level crop-yield data; Used as a baseline for PS-07 counterfactual/yield modelling.
ICRISAT VDSA	Historical district-level agricultural context, especially for rainfed and semi-arid crops.
Agmarknet mandi-price data	Current and historical wholesale prices for PS-03 economic optimization and ROI.
Open-access FAO / ICRISAT agronomic literature	Crop-specific agronomic thresholds for heat, frost, waterlogging, GDD and phenology.

Additional resources are limited to specific modelling requirements where an equivalent organizer-provided variable is not available.

05) The Technology Stack:

Our system is designed as a Google Cloud–native architecture, with Google Cloud services forming the core of its AI, voice, data and ML infrastructure. Rather than adding cloud services as isolated components, each is tied to a specific system requirement from zero-friction farmer onboarding and vernacular interaction to offline resilience and model execution.

Figure 3 shows the farmer interface to the multimodal AI layer, backend intelligence, data infrastructure and ML pipelines.

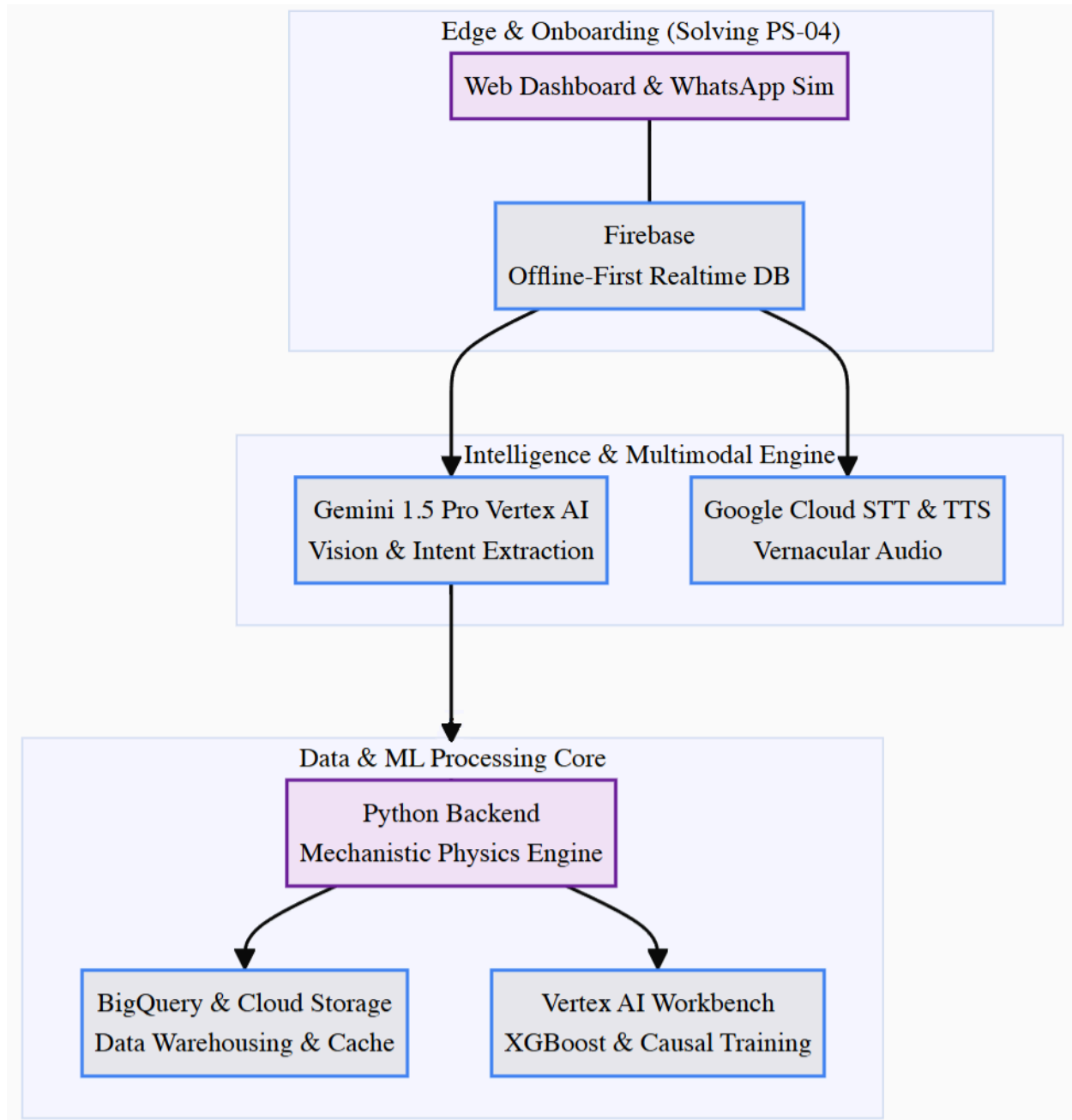


Figure 3. Google Cloud Architecture Flow

05.1) Core Technology Stack:

Layer	Technology	Role in The Solution
Farmer Interface	Web Dashboard - WhatsApp-style UI	Provides a lightweight farmer-facing interaction layer.
AI & Multimodal	Gemini + Google STT/TTS	Extracts intent from vernacular voice, supports multilingual interaction and provides vision-based fallback.
Application & Physics	Python + FastAPI	Runs the backend, mechanistic agronomic calculations and data orchestration.
ML & Causal Inference	Scikit-learn, XGBoost, EconML	Supports weather-yield modelling and Double ML treatment-effect estimation.
Cloud & Data	Firebase, BigQuery, Cloud Storage, Vertex AI	Provides persistence, data processing, model execution and storage infrastructure.
Agronomic APIs	Meteoblue + CE Hub	Supply the organiser-provided weather, soil and agronomic inputs used by the intelligence pipeline.

06) Demonstrate Feasibility:

Our solution is feasible within the Build Sprint because we are not developing new science from scratch; we are integrating established models, APIs, and ML components into one modular pipeline. We have already implemented and tested some of the ML models shared in our earlier document, giving us a working foundation to build on.

The core agronomic calculations such as GDD, VPD, and FAO Penman-Monteith are established methods that can be implemented directly in Python without heavy training or GPU requirements. For stress detection, we combine existing CE Hub outputs with our own stage-aware stress indices to create a more useful crop-level signal. The multilingual layer uses existing speech-to-text, LLM, and text-to-speech APIs, making voice interaction practical to implement during the sprint.

For yield attribution, we use XGBoost with historical yield and weather data, along with established causal-inference libraries such as EconML. These are lightweight, ready-to-use components rather than models we need to build from scratch.

Most importantly, the architecture is modular: stress detection, recommendations, multilingual interaction, and yield attribution can be developed independently and connected progressively. We also have a fallback through Open-Meteo if any external weather API becomes unavailable, and demo data can be cached to avoid dependency on live API calls. This makes the overall system realistic to develop, test, and demonstrate within the hackathon timeline.

06.1) What already exists vs what we build:

Already exists / validated	We build during hackathon	Future extension
GDD, VPD, FAO ET formulas (textbook science)	Weather-to-stress pipeline + Biological Clock	Multi-season feedback learning loop
CE Hub + Meteoblue APIs (tested, documented)	Product recommendation engine + dosage slider	NPK sensor integration
Gemini 1.5 Pro multilingual (API ready)	WhatsApp-style voice interface	District-wide spatial risk mapping
XGBoost + EconML libraries (pip install)	ROBI calculator + What-If simulator	Production-scale deployment
Syngenta product knowledge (from provided data)	End-to-end demo with Ramesh storyline	Retailer-facing supply forecast dashboard

07) Expected Impact:

The solution aims to help farmers make timely, agronomically sound and financially informed crop-input decisions, replacing guesswork and routine-based application with field-specific recommendations.

The system identifies crop stress based on growth stage and local conditions, then translates it into a practical recommendation covering **product, dosage, timing and confidence**. It also compares the financial implications of applying, delaying or skipping the intervention using expected yield impact and prevailing Mandi prices, while delivering the recommendation through a **regional-language voice advisory**. After harvest, the system estimates the counterfactual yield, measures the treatment effect and calculates the resulting **revenue impact and ROBI**, providing a measurable view of the decision's value.

07.1) Expected Outcomes:

- 3–14 days of advance stress visibility, enabling action before visible crop damage.
- Crop and condition-specific product and dosage recommendations, based on growth stage, weather and field conditions.
- Reduced unnecessary input use through dosage optimisation and economic checks.
- Apply / Delay / Skip comparison using expected yield impact and prevailing Mandi prices, expressed as (₹/acre).
- Regional-language voice advisories that make recommendations easier to understand and act upon.
- Post-harvest impact measurement, including estimated yield gain, revenue gain and Return on Biological Investment (ROBI).

For Syngenta and other agricultural stakeholders, the system can build a better understanding of which products work, where they work, at which crop stages, and under what weather and field conditions, supporting better product targeting, demand planning and evidence-based farmer engagement. More broadly, it makes crop-input decisions more timely, informed and financially transparent, helping farmers understand not only what action to take, but whether that action is actually worth the money they are spending.

07.2) For Syngenta:

- Evidence-based product adoption: Demonstrates product value through measurable treatment outcomes rather than relying solely on conventional product communication.
- Zone- and condition-specific efficacy intelligence: Captures relationships between product performance, crop stage, soil and weather conditions.
- Proactive demand planning: Advance stress visibility can help anticipate regional product-demand patterns.
- Wider farmer accessibility: Voice-first regional-language interaction lowers the usability barrier for farmers less comfortable with text-based interfaces.

08) Team contribution:

Team Members	Role in the Team
Divyansh Sharma (Team Lead)	System Design and ML Architecture
Ritvik Malhotra	Agronomic Intelligence & Decision Logic
Ishaan Sen	Full Stack Developer
Sameer Mishra	Backend Engineering & API Integration
Rishabh Barthwal	Machine Learning & Data Analytics

Github Link- <https://github.com/23f2003927/hyperion>